STAT 516: Discrete-Time Markov Chains

Liang-Cheng Chen

Autumn 2026

Reading guide

English transcription of `notes/stat516-note3.pdf`, updated October 26, 2026 (classes of October 20 and 22). Checked against Yen-Chi Chen's supplied *Lecture 3: Discrete-Time Markov Chain – Part I* (L3), Sections 3.1–3.6. Source-page labels refer to the handwriting; page 12 is blank. Yellow boxes retain both explicit TODOs with proposed completions; red boxes explain corrections; blue boxes add lecture-based context.

1 Stochastic processes and coin tossing

Handwritten source: 1; L3, Section 3.1.

A stochastic process is a family of random variables indexed by time. In discrete time we write $\{X_n : n = 0, 1, 2, ...\}$, with X_n taking values in a state space S. A continuous index set instead gives a continuous-time process. Discrete time does not require discrete-valued states. Here we focus on finite or countably infinite state spaces, which can be labeled by integers.

Let $Y_1, Y_2, ...$ be IID $\text{Ber}(\theta)$, with heads encoded as 1. Define

$$H_n = \sum_{r=1}^{n} Y_r, \qquad \overline{Y}_n = H_n/n \quad (n \geq 1), \tag{1}$$

$$Z_n = \sum_{r=1}^{n} (\mathbf{1}_{\{Y_r=1\}} - \mathbf{1}_{\{Y_r=0\}}) = 2H_n - n. \tag{2}$$

Here H_n counts heads, $\overline{Y}_n$ is their proportion, and Z_n is the excess of heads over tails. Set $H_0 = Z_0 = 0$. The updates are

$$H_{n+1} = H_n + Y_{n+1}, \tag{3}$$

$$Z_{n+1} = Z_n + 2Y_{n+1} - 1. \tag{4}$$

Given $H_5 = 3$, $H_6 = 4$ with probability θ and $H_6 = 3$ with probability $1 - \theta$. Both processes are generally dependent over time despite the independent tosses.

Correction: Count versus proportion (source p. 1)

The handwritten $n^{-1} \sum_r Y_r$ is a proportion, not the number of heads. The example $X_5 = 3$ and the update by 0 or 1 concern the count H_n. We retain both statistics with distinct notation. The proportion is also Markov, but its update

$$\overline{Y}_{n+1} = \frac{n\overline{Y}_n + Y_{n+1}}{n+1} \tag{5}$$

depends explicitly on time; the count and excess processes have time-homogeneous transition rules.

2 The Markov property and transition matrices

Handwritten source: 2–3; L3, Section 3.2.

A process is a Markov chain if, for $n \geq 1$ and histories of positive probability,

$$\mathbb{P}(X_n = j \mid X_{n-1} = i, \dots, X_0 = x_0) = \mathbb{P}(X_n = j \mid X_{n-1} = i). \tag{6}$$

The distribution of the next state given the past depends only on the present state. Both H_n and Z_n satisfy this condition. Markov processes appear in MCMC and in suitable formulations of SGD, Langevin dynamics, and diffusion models.

Correction: Condition on the past (source p. 2)

The shorthand $p(X_n \mid X_{-n}) = p(X_n \mid X_{n-1})$ is misleading if X_{-n} denotes all other variables, including future observations. The definition conditions only on the past; the future can provide additional information about X_n.

A chain is *time homogeneous* if each one-step transition probability is independent of time. With rows indexing the current state and columns the next state, write

$$p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i), \qquad P = (p_{ij}). \tag{7}$$

Time homogeneity means constancy over n for each fixed pair (i, j), not equality of all entries. A transition kernel can be specified even at states with zero probability under the chosen initial law.

For $S = \{1, \dots, s\}$, P is an $s \times s$ row-stochastic matrix:

$$p_{ij} \geq 0, \qquad \sum_{j=1}^{s} p_{ij} = 1 \quad \text{for every } i. \tag{8}$$

#TODO: Why is the all-ones vector an eigenvector? (source p. 3)

Let $\mathbf{1}_s = (1, \dots, 1)^T$. Each component of the product is a row sum:

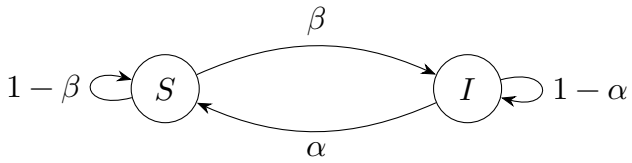
$$(P\mathbf{1}_s)_i = \sum_{j=1}^{s} p_{ij} = 1 = (\mathbf{1}_s)_i. \tag{9}$$

Thus $P\mathbf{1}_s = \mathbf{1}_s$: the nonzero vector $\mathbf{1}_s$ is a right eigenvector with eigenvalue 1. A stationary distribution, by contrast, is a normalized left eigenvector.

2.1 SIS model

Observe an individual's status: S (susceptible) or I (infected). Let β be the infection probability and α the recovery probability in one step. In state order (S, I),

$$P = \begin{pmatrix} 1-\beta & \beta \\ \alpha & 1-\alpha \end{pmatrix}, \qquad 0 \le \alpha, \beta \le 1. \tag{10}$$



L3, Example 3.3 initially lists states in reverse order (I, S). This matrix follows the handwriting; reversing the order permutes both rows and columns.

3 Ehrenfest diffusion and transition diagrams

Handwritten source: 4–5; L3, Example 3.4.

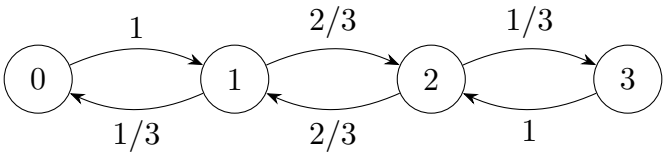
A box contains $N \ge 1$ particles in two compartments. At each discrete step, choose one particle uniformly and move it to the other compartment. Let X_n be the number in compartment 1 after n moves. The state space is $\{0, \dots, N\}$. Given $X_n = i$, moving a particle out of compartment 1 decreases the count, whereas moving one out of compartment 2 increases it:

$$p_{ij} = \begin{cases} i/N, & j = i-1, \\ 1 - i/N, & j = i+1, \\ 0, & \text{otherwise}, \end{cases} \qquad i, j \in \{0, \dots, N\}. \tag{11}$$

In particular, $p_{0,1} = p_{N,N-1} = 1$. The time index counts moves, rather than elapsed

physical time between moves. For $N = 3$, in order $(0, 1, 2, 3)$,

$$P = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1/3 & 0 & 2/3 & 0 \\ 0 & 2/3 & 0 & 1/3 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{12}$$



A transition diagram is a weighted directed graph. Its vertices are the states, and an arrow $i \to j$ of weight $p_{ij} > 0$ indicates a possible one-step move. Outgoing weights at each vertex sum to 1. This graph may contain cycles and self-loops.

Supplement: A useful two-step check

Starting at 0, the first move goes to 1 with certainty. The second move returns to 0 with probability 1/3 or goes to 2 with probability 2/3. Thus

$$\mathbb{P}(X_2 = 0 \mid X_0 = 0) = 1/3, \qquad \mathbb{P}(X_2 = 2 \mid X_0 = 0) = 2/3. \tag{13}$$

This illustrates why multi-step probabilities must sum over intermediate paths.

4 Joint probabilities and multi-step memory

Handwritten source: 6–7; L3, Section 3.3.

4.1 Path probability and transition counts

Let $\mu_i = \mathbb{P}(X_0 = i)$ be the initial distribution. The chain rule and Markov property give

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{r=1}^{n} p(x_r \mid x_{r-1}, \dots, x_0) \tag{14}$$

$$= \mu_{x_0} \prod_{r=1}^{n} p_{x_{r-1}, x_r}. \tag{15}$$

Thus a homogeneous chain is specified by its initial law and transition kernel. For a finite state space, grouping equal transitions yields

$$N_{ij} = \sum_{r=1}^{n} \mathbf{1}_{\{x_{r-1}=i,\ x_r=j\}}, \tag{16}$$

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{i,j \in S} p_{ij}^{N_{ij}}. \tag{17}$$

Here N_{ij} counts transitions from i to j. Factors with $N_{ij} = 0$ are interpreted as 1; any path containing an impossible transition has probability zero.

4.2 The latest observation is sufficient for prediction

For $m \geq 1$, the one-step Markov condition implies

$$\mathbb{P}(X_{n+m} = j \mid X_n = i, \dots, X_0 = x_0) = \mathbb{P}(X_{n+m} = j \mid X_n = i). \tag{18}$$

In the handwritten example ($n = 1, m = 2$), sum over the unobserved intermediate state:

$$p(x_3 \mid x_1, x_0) = \sum_{x_2} p(x_3 \mid x_2, x_1, x_0) p(x_2 \mid x_1, x_0) \tag{19}$$

$$= \sum_{x_2} p(x_3 \mid x_2) p(x_2 \mid x_1) = p(x_3 \mid x_1). \tag{20}$$

The last equality uses the following marginalization identity, also derived in the handwriting:

$$p(x_3 \mid x_2, x_1) = \sum_{x_0} p(x_3 \mid x_2, x_1, x_0) p(x_0 \mid x_2, x_1) \tag{21}$$

$$= p(x_3 \mid x_2) \sum_{x_0} p(x_0 \mid x_2, x_1) = p(x_3 \mid x_2). \tag{22}$$

All conditional identities apply only where the conditioning events have positive probability; zero-weight summands are omitted.

Supplement: Past and future are independent given the present

The path factorization also implies, for finite past and future blocks,

$$(X_0, \dots, X_{n-1}) \perp\!\!\!\perp (X_{n+1}, \dots, X_{n+m}) \mid X_n. \tag{23}$$

Conditioning on the present separates the past factors from the future factors, connecting the chain to the graphical models in Notes 1 and 2.

5 Multi-step transitions

Handwritten source: 8–9; L3, Section 3.4, Lemmas 3.6–3.7.

For a homogeneous chain define

$$p_{ij}^{(n)} = \mathbb{P}(X_n = j \mid X_0 = i), \qquad p_{ij}^{(0)} = \mathbf{1}_{\{i=j\}}. \tag{24}$$

The zero-step kernel is the identity. Time homogeneity implies, for any $r \geq 0$,

$$\mathbb{P}(X_{r+n} = j \mid X_r = i) = p_{ij}^{(n)}. \tag{25}$$

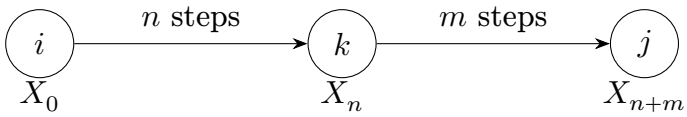
To see this, expand over all intermediate paths. The product of the n one-step transition probabilities depends on the path, not its starting time.

5.1 Chapman–Kolmogorov equation

For integers $n, m \geq 0$,

$$p_{ij}^{(n+m)} = \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}. \tag{26}$$

Choose an intermediate state k after the first n steps, then sum over all its possible values:



#TODO: Supply the proof (source p. 9)

Condition on X_n, apply the multi-step Markov property, and then use time homogeneity:

$$\begin{aligned} p_{ij}^{(n+m)} &= \sum_{k \in S} \mathbb{P}(X_{n+m} = j, X_n = k \mid X_0 = i) && (27)\\ &= \sum_{k \in S} \mathbb{P}(X_{n+m} = j \mid X_n = k, X_0 = i)\mathbb{P}(X_n = k \mid X_0 = i) && (28)\\ &= \sum_{k \in S} \mathbb{P}(X_{n+m} = j \mid X_n = k) p_{ik}^{(n)} && (29)\\ &= \sum_{k \in S} p_{ik}^{(n)} p_{kj}^{(m)}. && (30) \end{aligned}$$

Terms with $p_{ik}^{(n)} = 0$ contribute zero without needing a conditional probability on an impossible event. For $n = 0$ or $m = 0$, the formula follows from the identity kernel.

5.2 Forward and backward equations

Taking one of the two increments to be 1 gives

$$p_{ij}^{(n+1)} = \sum_{k\in S} p_{ik}^{(n)} p_{kj} \qquad \text{(forward: single out the final step)}, \tag{31}$$

$$p_{ij}^{(n+1)} = \sum_{k\in S} p_{ik} p_{kj}^{(n)} \qquad \text{(backward: single out the first step)}. \tag{32}$$

6 Matrix powers and marginal laws

Handwritten source: 9–10; L3, Section 3.4 and Examples 3.8–3.9.

Writing $P^{(n)} = (p_{ij}^{(n)})$, Chapman–Kolmogorov becomes

$$P^{(n+m)} = P^{(n)} P^{(m)}, \tag{33}$$

$$P^{(n+1)} = P^{(n)} P = P P^{(n)}, \qquad P^{(0)} = I, \tag{34}$$

$$P^{(2)} = P^2, \quad P^{(3)} = P^3, \quad \dots, \quad P^{(n)} = P^n. \tag{35}$$

Matrix powers are not entrywise powers (source p. 10)

The identity is $p_{ij}^{(n)} = (P^n)_{ij}$, generally not $(p_{ij})^n$. For the $N = 3$ Ehrenfest chain, $p_{00} = 0$, but

$$p_{00}^{(2)} = p_{01}p_{10} = 1/3 \neq (p_{00})^2 = 0. \tag{36}$$

Supplement: Evolving the marginal distribution

With the initial law written as a row vector $\mu = (\mu_i)$,

$$\mu^{(n)} = \mu P^n, \qquad \mathbb{P}(X_n = j) = \sum_i \mu_i p_{ij}^{(n)}. \tag{37}$$

For column-vector distributions the corresponding rule is multiplication by $(P^T)^n$ on the left.

6.1 When identical rows give IID observations

If every row of the one-step matrix equals a probability vector q, then

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \mu_{x_0} \prod_{r=1}^{n} q_{x_r}. \tag{38}$$

Thus $X_1, X_2, \ldots$ are IID with law q and independent of X_0. Including X_0 in the IID sequence additionally requires $\mu = q$. For the handwritten SIS order (S, I), encode $S = 0$, $I = 1$. If $\beta = 1 - \alpha$, then

$$P = \begin{pmatrix} \alpha & 1-\alpha \\ \alpha & 1-\alpha \end{pmatrix}, \tag{39}$$

so $X_1, X_2, \ldots$ are IID $\mathrm{Ber}(1-\alpha)$.

Correction: Distinguish one-step and long-lag independence (source p. 10)

Identical rows of P give the IID conclusion from time 1. Identical rows of a particular P^n instead imply $X_n \perp\!\!\!\perp X_0$; they do not make consecutive states IID. A limit with identical rows describes long-lag behavior, not exact finite-lag independence. Such a limit need not exist: the powers of $P = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ alternate between P and I (L3, Example 3.9).

7 First-step analysis: gambler's ruin

Handwritten source: 11; L3, Section 3.5, Example 3.10.

A gambler starts with a dollars and stops upon reaching either 0 or $M = a + b$, with a, b positive integers. Each round independently increases the fortune by 1 with probability p or decreases it by 1 with probability $q = 1 - p$. The aim is total wealth $a + b$, equivalently a gain of b dollars from the initial a. With $X_0 = a$ and $S = \{0, \ldots, M\}$, the transition probabilities are

$$p_{i,i+1} = p, \quad p_{i,i-1} = q \qquad (1 \le i \le M-1), \tag{40}$$

$$p_{00} = p_{MM} = 1. \tag{41}$$

All other entries are zero. Let

$$T = \inf\{n \ge 0 : X_n \in \{0, M\}\} \tag{42}$$

be the absorption time.

Supplement: Complete the first-step calculation from the lecture

The handwriting gives the setup only. Define the probability of reaching the upper boundary by

$$u_i = \mathbb{P}(X_T = M \mid X_0 = i), \qquad u_0 = 0, \quad u_M = 1. \tag{43}$$

Conditioning on the first move yields

$$u_i = p u_{i+1} + q u_{i-1}, \qquad 1 \le i \le M-1. \tag{44}$$

For $0 < p < 1$, set $d_i = u_i - u_{i-1}$. Since $p + q = 1$,

$$p(u_{i+1} - u_i) = q(u_i - u_{i-1}), \qquad d_{i+1} = \frac{q}{p} d_i. \tag{45}$$

Put $r = q/p$. Telescoping and the two boundaries give

$$u_i = d_1 \sum_{k=0}^{i-1} r^k, \qquad 1 = u_M = d_1 \sum_{k=0}^{M-1} r^k. \tag{46}$$

Therefore the desired probability from the initial fortune is

$$u_a = \begin{cases} \dfrac{a}{a+b}, & p = q = 1/2, \\ \dfrac{1 - (q/p)^a}{1 - (q/p)^{a+b}}, & 0 < p < 1,\ p \neq q. \end{cases} \tag{47}$$

At the endpoints, $u_a = 0$ for $p = 0$ and $u_a = 1$ for $p = 1$. First-step analysis conditions on the first transition, restarts from the resulting state, and solves with the boundary conditions.

8 Accessibility and communicating classes

Handwritten source: 13–14; L3, Section 3.6, Lemma 3.12.

State j is *accessible from* i, written $i \to j$, if

$$p_{ij}^{(n)} > 0 \quad \text{for some integer } n \geq 0. \tag{48}$$

The path need not have only one step. The $n = 0$ convention ensures $i \to i$ even without a later positive-probability return. States *communicate*, written $i \leftrightarrow j$, if both $i \to j$ and $j \to i$.

8.1 Communication is an equivalence relation

$$i \leftrightarrow i \qquad \text{(reflexivity)}, \tag{49}$$

$$i \leftrightarrow j \implies j \leftrightarrow i \qquad \text{(symmetry)}, \tag{50}$$

$$i \leftrightarrow j, \quad j \leftrightarrow k \implies i \leftrightarrow k \qquad \text{(transitivity)}. \tag{51}$$

Reflexivity follows from $P^{(0)} = I$; symmetry is part of the definition. For transitivity of accessibility, choose n, m with $p_{ij}^{(n)} > 0$ and $p_{jk}^{(m)} > 0$. Then

$$p_{ik}^{(n+m)} = \sum_{\ell \in S} p_{i\ell}^{(n)} p_{\ell k}^{(m)} \geq p_{ij}^{(n)} p_{jk}^{(m)} > 0. \tag{52}$$

Apply the same argument in reverse for communication. The equivalence classes are *communicating classes*. A chain with a single class is *irreducible*.

Correction: Accessibility need not be symmetric (source p. 14)

The handwritten symmetry line uses $i \to j \Rightarrow j \to i$. This is false for accessibility: entering an absorbing state may leave no return path. The symmetric relation in the lemma must be communication $\leftrightarrow$. Accessibility is reflexive and transitive; communication has all three equivalence properties.

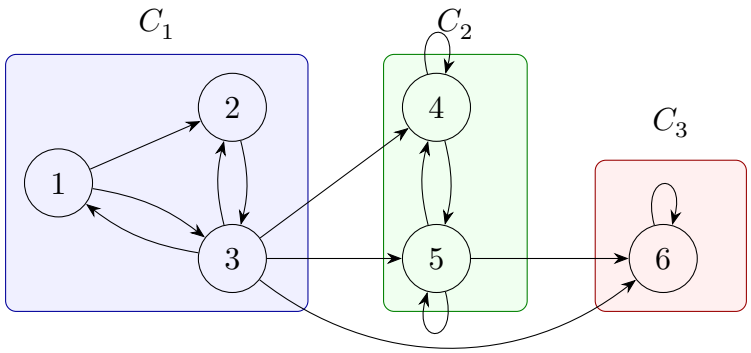
8.2 The six-state example

Handwritten source: 13; compare L3, Section 3.6.

The handwritten graph has three communicating classes:

$$C_1 = \{1, 2, 3\}, \qquad C_2 = \{4, 5\}, \qquad C_3 = \{6\}. \tag{53}$$

Within C_1, arrows are $1 \to 2$, $1 \leftrightarrow 3$, and $2 \leftrightarrow 3$. Additional arrows are $3 \to 4$, $3 \to 5$, $3 \to 6$, $4 \leftrightarrow 5$, and $5 \to 6$, with self-loops at 4, 5, and 6. No numerical weights are specified.



Each arrow means a strictly positive one-step probability, not necessarily probability 1. Outgoing weights must sum to 1. State 6 has only a self-loop, so it is absorbing with $p_{66} = 1$.

Supplement: Topological order applies to the classes

The handwritten keyword "topological ordering" applies after collapsing each communicating class to one vertex. The class graph has edges

$$C_1 \longrightarrow C_2, \qquad C_1 \longrightarrow C_3, \qquad C_2 \longrightarrow C_3. \tag{54}$$

It is acyclic: a cycle between distinct classes would make them mutually reachable, merging them into one class. Thus (C_1, C_2, C_3) is a topological order, although the original graph contains cycles.

A class is *closed* if no transitions leave it. Here only C_3 is closed. Since this chain is finite, C_1 and C_2 are transient classes; C_3 is recurrent and absorbing. This classification is a lecture-based consequence beyond the handwritten accessibility definitions.

Ordering the states by these classes produces a block upper-triangular matrix:

$$P = \begin{pmatrix} P_{11} & P_{12} & P_{13} \\ 0 & P_{22} & P_{23} \\ 0 & 0 & 1 \end{pmatrix}. \tag{55}$$

The diagonal blocks describe moves within a class. The blocks P_{11} and P_{22} need not themselves be stochastic, because probability can leave those classes.